3-LABORATORIYA ISHI

Mavzu: Ma’lumotlarni vizualizatsiya qilish (Matplotlib va Seaborn)

Maqsad

Ushbu laboratoriya ishining maqsadi:

- Grafiklar yordamida ma’lumotlarni tahlil qilish
- Scatter, Histogram, Bar chart, Heatmap chizish
- Korrelyatsiyani aniqlash
- Natijalarni tahlil qilish

Kerakli vositalar

- Python 3.10+
- Pandas
- Matplotlib
- Seaborn

O‘rnatish:

```
pip install pandas matplotlib seaborn
```

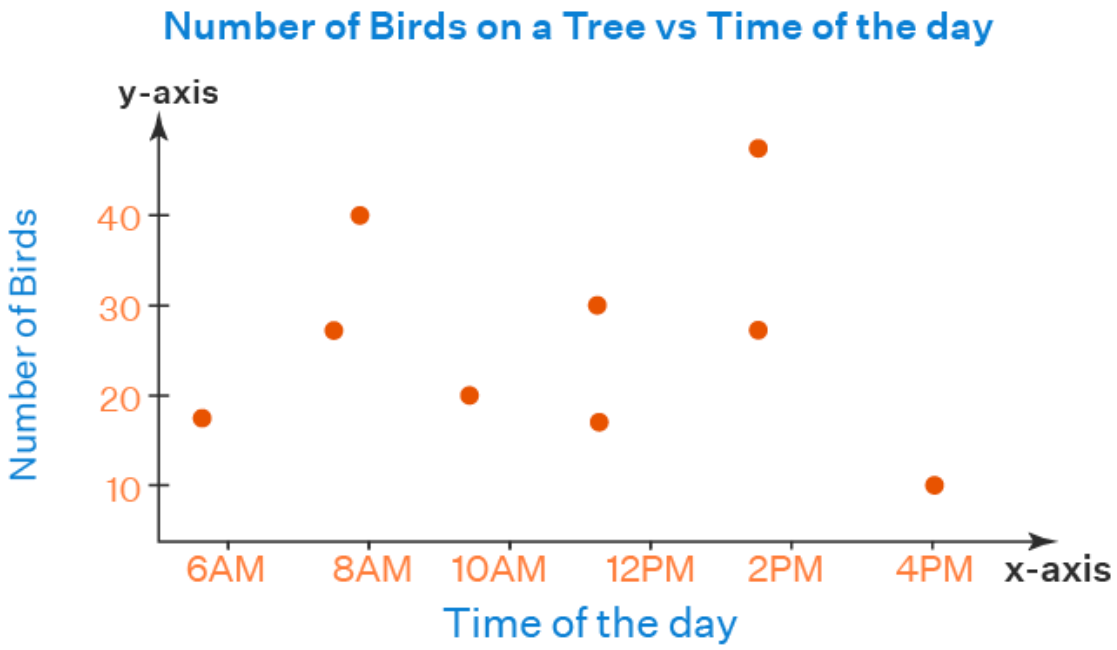
Nazariy asos

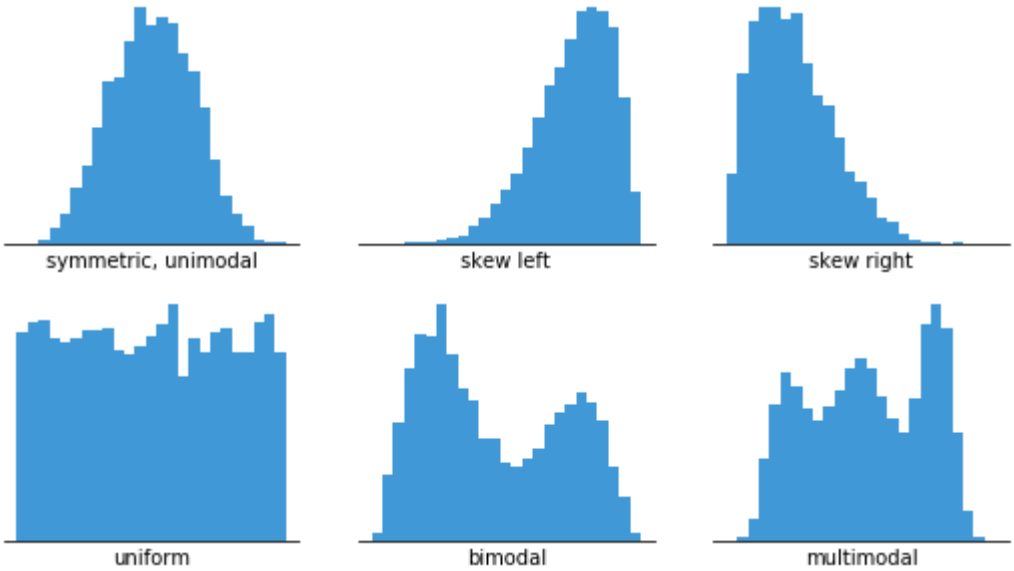
Vizualizatsiya AI jarayonida juda muhim bosqich. Grafiklar orqali:

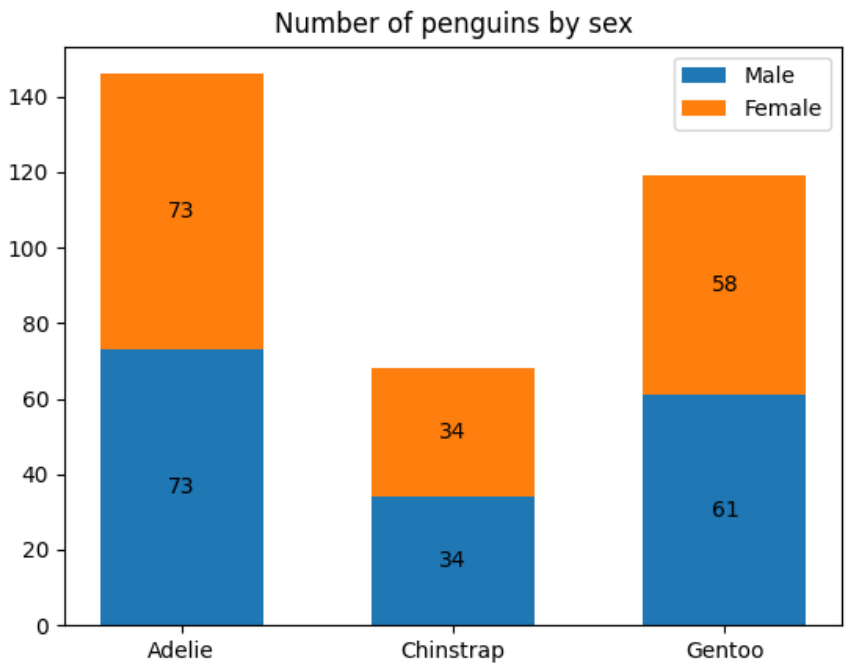
- Ma’lumot taqsimoti
- Bog‘liqlik (correlation)
- Trend
- Outlier

aniqlanadi.

Grafik turlari


Number of Birds on a Tree vs Time of the day
y-axis
Number of Birds
40
30
20
10
6AM
8AM
10AM
12PM
2PM
4PM
x-axis
Time of the day


symmetric, unimodal
skew left
skew right
uniform
bimodal
multimodal



Ishlash tartibi (Step-by-Step)

1-qadam: Dataset yuklash

students.csv (2-laboratoriyadagi fayl)

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

df = pd.read_csv("students.csv")
```

2-qadam: Scatter plot (Math vs Physics)

```
plt.scatter(df["math"], df["physics"])
plt.xlabel("Math")
plt.ylabel("Physics")
plt.title("Math va Physics bog‘liqligi")
plt.show()
```

3-qadam: Histogram (Math ball taqsimoti)

```
plt.hist(df["math"], bins=5)
plt.title("Math ball taqsimoti")
plt.show()
```

4-qadam: Bar chart (Gender bo‘yicha o‘rtacha ball)

```
df["gender"] = df["gender"].map({"M":0,"F":1})

group = df.groupby("gender")["math"].mean()

group.plot(kind="bar")
plt.title("Gender bo‘yicha Math o‘rtacha ball")
plt.show()
```

5-qadam: Correlation Heatmap

```
sns.heatmap(df.corr(numeric_only=True), annot=True, cmap="coolwarm")
plt.title("Korrelyatsiya matritsasi")
plt.show()
```

To‘liq kod (Yakuniy variant)

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

df = pd.read_csv("students.csv")

# Scatter
plt.scatter(df["math"], df["physics"])
plt.xlabel("Math")
plt.ylabel("Physics")
plt.title("Math vs Physics")
plt.show()

# Histogram
plt.hist(df["math"], bins=5)
plt.title("Math Distribution")
plt.show()

# Bar chart
df["gender"] = df["gender"].map({"M":0,"F":1})
df.groupby("gender")["math"].mean().plot(kind="bar")
plt.title("Gender vs Math")
plt.show()

# Heatmap
sns.heatmap(df.corr(numeric_only=True), annot=True, cmap="coolwarm")
plt.title("Correlation Matrix")
plt.show()
```

Talaba tekshiradi:

✅ 4 xil grafik chizildi
✅ Heatmap ko‘rsatildi
✅ Scatter plot chiqdi
✅ Grafiklar to‘g‘ri nomlandi

Hisobot topshirish talabi

Talaba topshiradi:

1. Python kodi
2. 4 ta grafik skrinshoti
3. 8-10 qatordan iborat tahlil:
    - Math va Physics bog‘liqligi qanday?
    - Qaysi fan kuchli korrelyatsiyaga ega?
    - Gender bo‘yicha farq bormi?

Mezoni	Ball
Grafiklar to‘g‘ri chiqishi	40
Heatmap tahlili	20
Kod tozaligi	15
Tahlil yozish	15
Grafik nomlanishi	10